

Clockster: An Analysis of Tardiness and Absences in a Medical Company

Problem

The HR Department in a medical company wants to know the following information about employee attendance and leaves:

- Identify the most disciplined and undisciplined employees and divisions
- Create a visualization with the analysis of weekdays and months when the most employees were late/absent (either for vacation or sick leave)
- Which heads of departments tend to forgive employees for lack of discipline? Are there any favorites for any heads of departments (perhaps some employees are always forgiven for being late, given time off, etc.)?

Methodology

The dataset comprised of the following:

- users - The list of all users (employees) and their basic information including the date of hire, date of leave, gender, department, position, etc.
- payroll - Contains information about the pay employees will receive for a particular time period.
- schedules - The schedules for the shifts of each employee. For this study, schedules pertaining to leaves and work are only considered.
- leave_requests - All requests for leaves. If approved, the requested leave will show up on the schedules dataset.
- attendance - The time in and time out recorded for each employee when reporting to work.

Using Excel, the case column in the attendance data was renamed to type to avoid problems in the later preprocessing stages via PostgreSQL. Then, using PostgreSQL, the following data preprocessing procedure was done:

- Cleaned the comma-separated text strings in the dates column (schedules and leave_requests), and in the user_id column (schedules), allowing for splitting and unpivoting of these columns later in Power Query.
- Concatenated the time_start and time_end columns with the timestamp column for schedules, and the date, time, and timestamp columns for attendance.
- Split the schedules data into work and leaves, and split the attendance data into in and out datasets.

- Saved the results to respective CSV files.

The CSV files are then imported into Power BI via Power Query, which involved the following procedures:

- Removal of duplicates:
 - Via user_id (users), user_id and date (leave_requests, schedules - leaves), user_id, date_start, and date_end (payroll)
 - Grouped by user_id and date and obtained the min timestamp (attendance - in) or max timestamp (attendance - out)
 - Grouped by user_id and date and obtained the min timestamp_start and max timestamp_end (schedules - work)
- Split by comma and unpivoted other columns for columns holding comma-separated values (schedules - work, schedules - leaves, and leave_requests)
- Full outer joined the schedules - work and schedules - leaves, and then merged the leave_requests (status), payroll, attendance - in, attendance - out, and users (created_at, date_hired, date_leave) data into it.

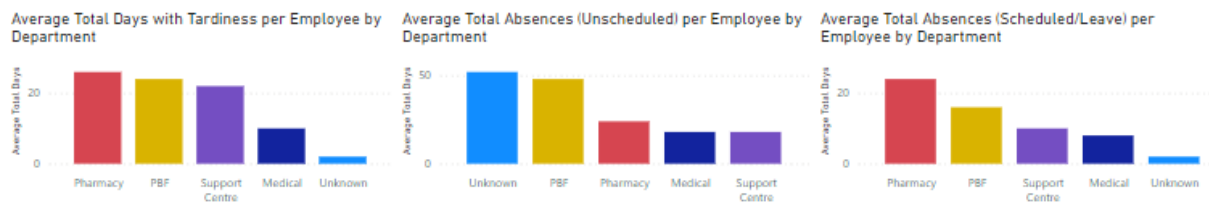
The following metrics were calculated using DAX:

- Calculated columns to indicate if a schedule record is:
 - Within employment period: Date should be between the date_hired (or created_at if date_hired is blank), inclusive, and date_leave (or none at all), exclusive.
 - Arrived late: The difference between time in and the scheduled time start in seconds.
 - Left early: The difference between scheduled time end and time out in seconds.
 - Basic Monthly Salary: Calculated based on the basic salary rate and the basic salary type (hourly, daily, or monthly).
- An employee is tardy for a particular day if a schedule record within the employment period has a time in and/or time out recorded and has arrived late by at least 10 minutes and/or left early. Schedule records with lates or early timeouts of more than 2 hours are considered outliers and are hence not considered tardy.
- An employee has an unscheduled absence for a particular day if a schedule record within the employment period has no time in and time out recorded and there is no leave type indicated.
- An employee has a scheduled/leave absence for a particular day if a schedule record within the employment period has no time in and time out recorded and there is a leave type indicated.
- For each employee, the number of distinct days wherein the employee was tardy or absent is calculated.
- For each department, the average number of distinct days with tardiness/absence per employee (for disciplined/undisciplined), the percentage of employees with tardiness/absence (for undisciplined forgiveness), and the percentage of approved leaves are calculated.
- For each month and weekday, the number of distinct employees with tardiness/absences are calculated.

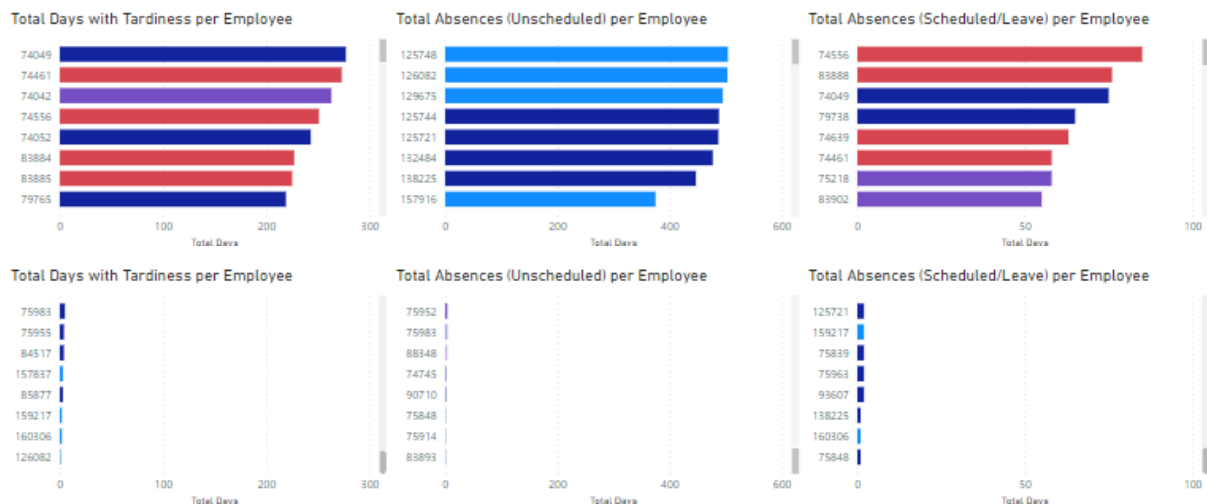
- Demographic information such as basic monthly salary and gender are also considered in the analysis of lateness and day-offs.

Findings

Identify the most disciplined and undisciplined employees and division



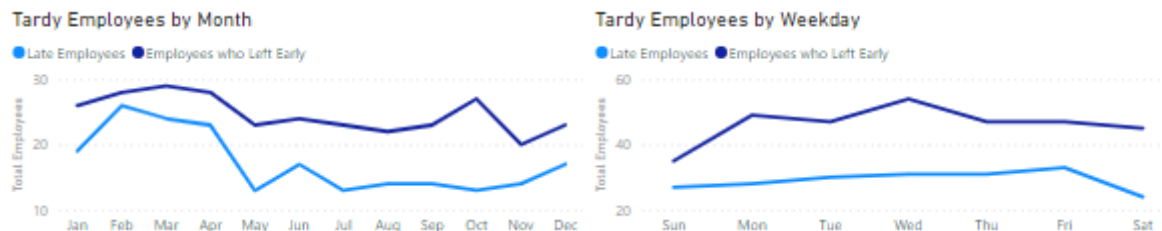
Based on the average total days with tardiness/absences per employee, for tardiness and scheduled/leave absences (i.e. Days with actual leaves filed for that day), Pharmacy is the most undisciplined while Unknown is the most disciplined. For unscheduled absences (i.e. Days scheduled as work days but the employee did not report to work as evidenced by the lack of time in and time out entries for the day despite a leave not filed for that day), Unknown is the most undisciplined while Support Centre and Medical are the most disciplined.



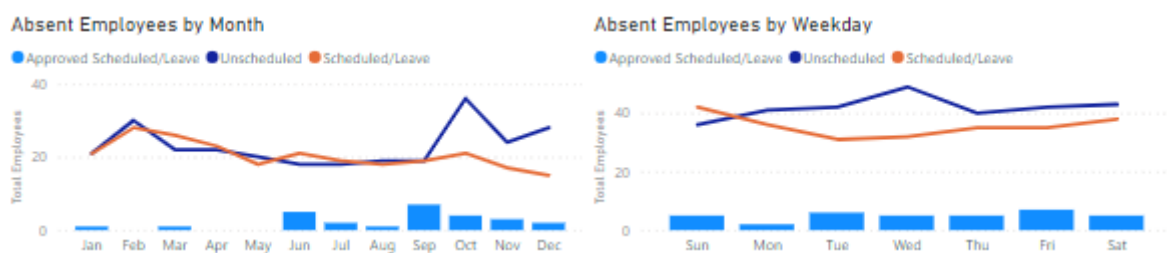
On a per-employee basis, in terms of tardiness, most of the undisciplined employees (i.e. employees with the highest number of days with tardiness) are from the Medical and Pharmacy departments, with some from Support Centre. Some Medical and Unknown department employees are the most disciplined (i.e. employees with the lowest number of days with tardiness). For Unscheduled Absences, the most undisciplined are the Unknown and Medical employees, while the most disciplined are from the Medical and Support Centre departments. For Scheduled Absences or Leaves, most of the undisciplined employees are from the Pharmacy, Support Centre, and Medical departments, while the most disciplined are mostly from the Medical and Unknown department. Very likely there are some

employees from the Unknown department who may not be familiar with the process of filing leaves.

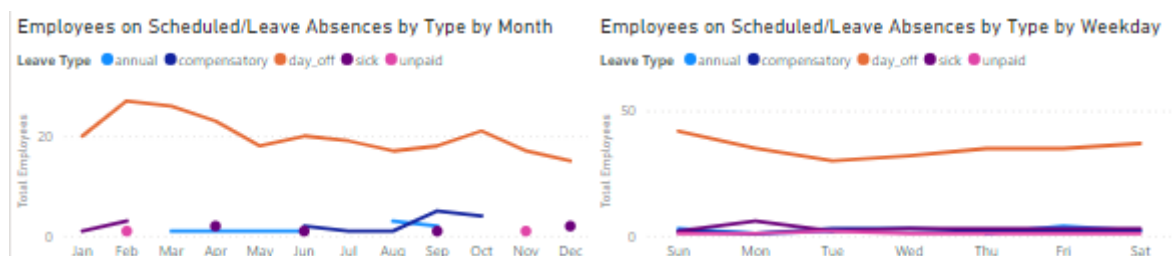
Create a visualization with the analysis of weekdays and months when the most employees were late/absent (either for vacation or sick leave)



February had the highest number of employees who are late or left early. Friday had the highest number of employees who are late while Wednesday had the highest number of employees who left early. There are more employees who left early than with lates.

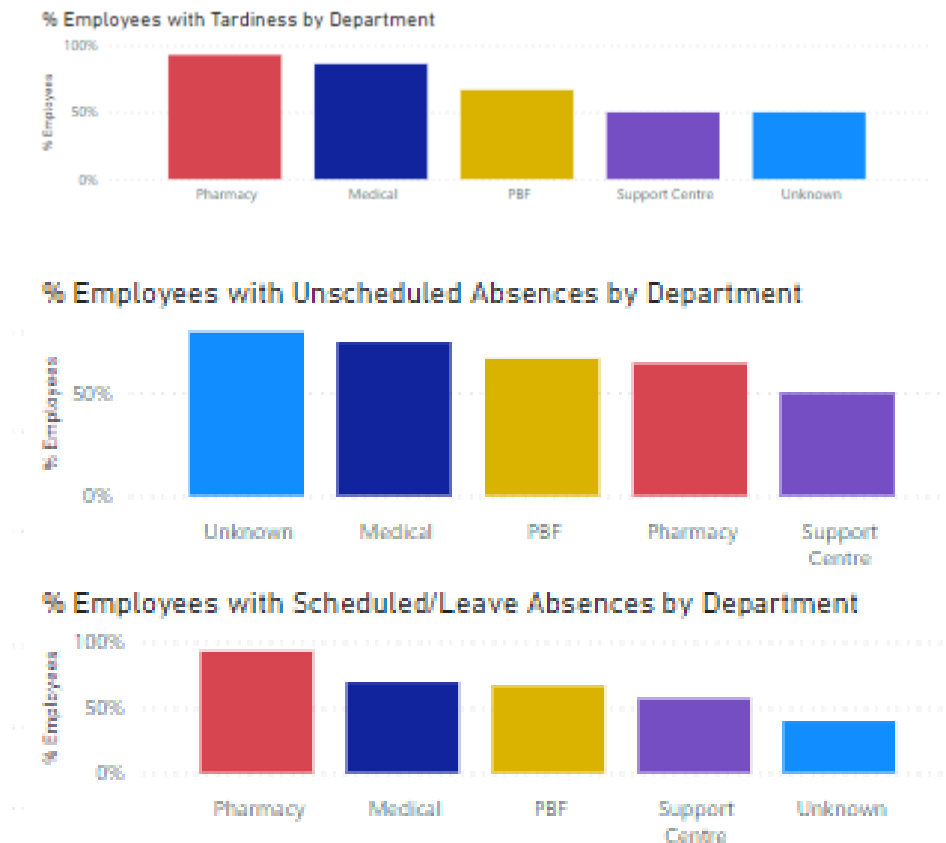


The highest number of employees with unscheduled absences occur in the month of October, while the highest number of employees with scheduled/leave absences occur in the month of February. Furthermore, September has the most number of employees with approved scheduled/leave absences. The highest number of employees with unscheduled absences is on Wednesday, while Sunday has the highest number of employees with scheduled/leave absences. Friday has the highest number of employees with approved scheduled/leave absences. There are only a handful of scheduled/leave absences actually marked as accepted or approved in the leave requests data.

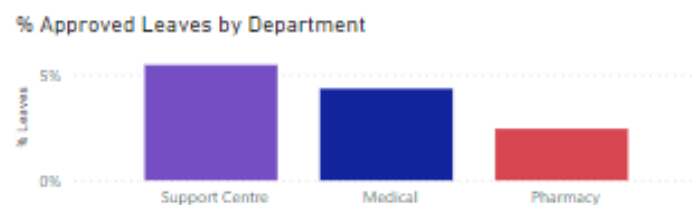


Most employees filed for day-off leaves. Annual/vacation leaves occur in the months of March to June, and August to September, while sick leaves are recorded in April, June, September, and December. Monday had the highest number of employees on sick leave, and Friday had the highest number of employees on annual/vacation leave.

Which heads of departments tend to forgive employees for lack of discipline? Are there any favorites for any heads of departments (perhaps some employees are always forgiven for being late, given time off, etc.)?



Pharmacy and Medical are the most forgiving in terms of tardiness, since there is a high percentage of employees in each department having a record of tardiness. Unknown and Medical are the most forgiving in terms of unscheduled absences, while in terms of scheduled/leave absences, Pharmacy and Medical are the most forgiving.



Support Centre has the highest percentage of approved leaves, and it should be noted that no approved leaves were recorded for PBF or Unknown departments. Even so, there are only a handful of approved leaves recorded.



Support Centre

% Employees with Lates

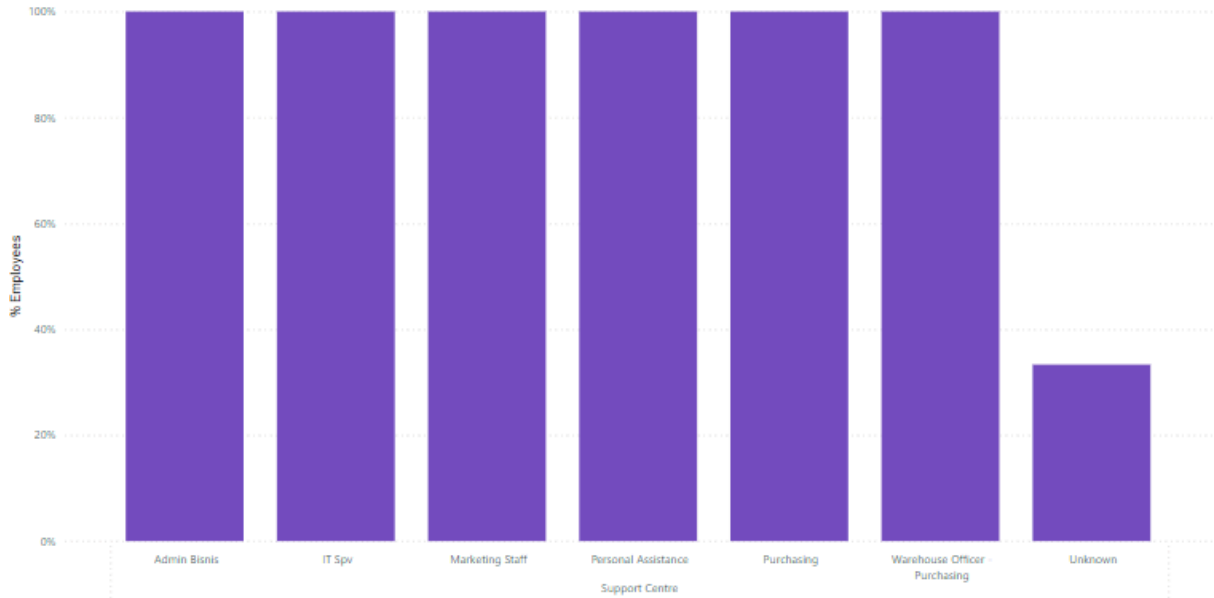


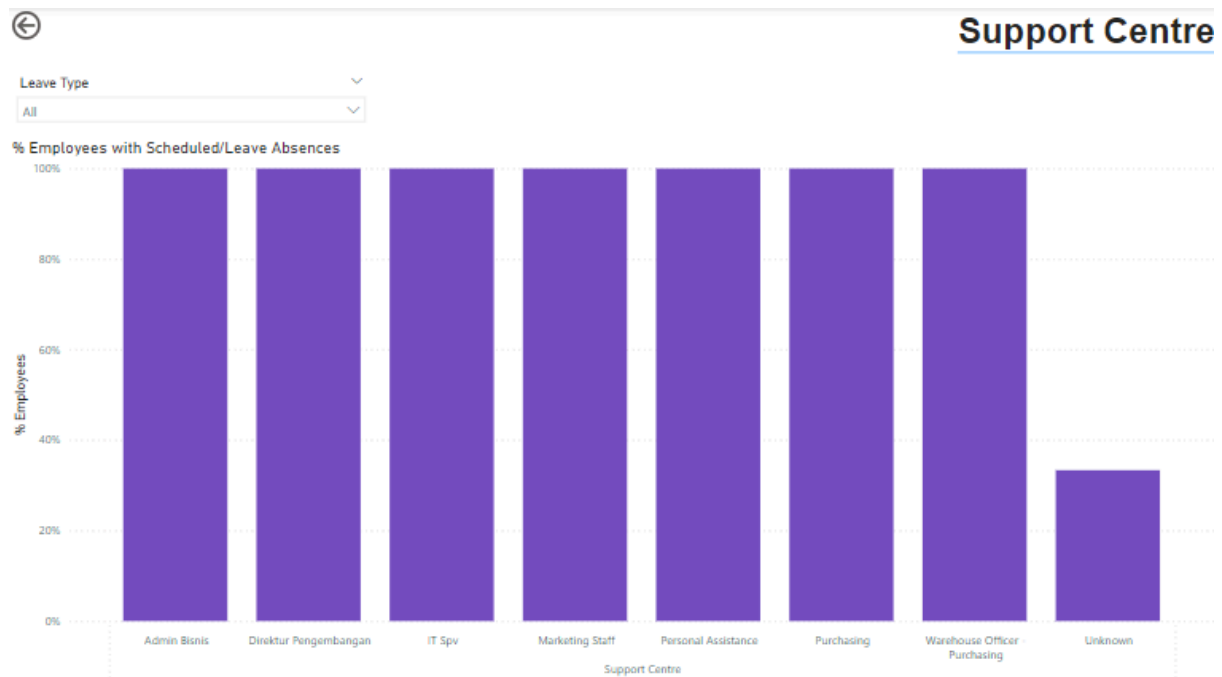
% Employees with Early Time-Outs



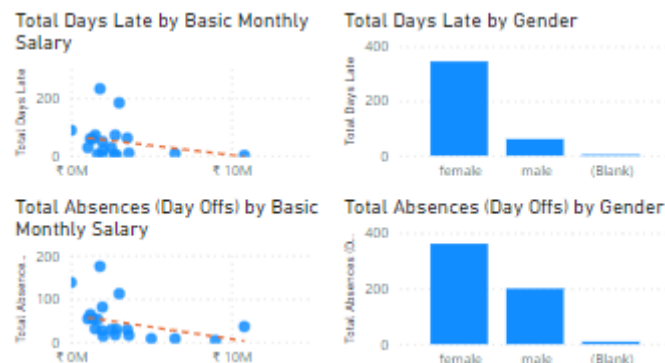
Support Centre

% Employees with Unscheduled Absences





From the drill-throughs on % Employees with Tardiness by Department, % Employees with Unscheduled Absences, and % Employees with Scheduled/Leave Absences by Department, Support Centre can be considered the most biased of the departments, as all except those with an Unknown position have a record of tardiness or absences.



In general, the higher the basic monthly salary, the less likely an employee will be late or file a day-off. Also, females are more likely to be late or file a day-off than males in general.

Recommendations

Based on the findings, the recommendations for the HR department are as follows:

- Employees in the Unknown department should be educated on how to properly file leaves.
- Encourage the Pharmacy, Medical, and Support Centre departments to enforce stricter guidelines in terms of tardiness and filing leaves, regardless of employee position.

- The fact that there are more employees who left early than employees being late may indicate potential burnout and should be further investigated.
- The fact that there are only a few recorded approved leaves might be an indication of inconsistency and inefficiency in the leave recording system and should be further investigated.
- Very likely employees only think that day-off is the only type of leave available. Employees should be made aware of the other leaves that are available to them. Alternatively, this could also indicate burnout and should be further investigated.
- One reason why employees with lower basic monthly salary are more likely to be late or file a day-off is because of employee demotivation. Efforts should be made to investigate the issue and offer salary raises if deemed necessary to motivate the employees.
- The fact that female employees have more days late or day-offs than males should be further investigated, since this may indicate a work-life balance issue.